Alireza Jafartash

Senior Back-End Software Engineer

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Summary

Back-end engineer with 4+ years in startups, building concurrency-safe ledgers, multi-tenant SaaS, high-throughput APIs, and real-time data pipelines across fintech, crypto, gaming, and retail. Also works on the product side: runs requirement sessions with owners who are not technical, turns them into specs, state machines, and release scope, and builds the front end for full-stack products. M.Sc. candidate in Computer Engineering, researching interpretability and hallucination in large language models.

Skills

- **Primary:** Go, TypeScript, Python, PostgreSQL, Redis, Apache Kafka
- **Frameworks:** Gin, Fiber (Go) · NestJS, Next.js (TypeScript) · Django, FastAPI (Python)
- **Familiar:** MongoDB, Kubernetes, Terraform, GCP, CI/CD, Monitoring (Prometheus, Grafana), UI/UX
- **Communication:** Explains systems clearly to engineers, non-technical owners, and AI coding agents through specs, decision records, and agent briefs. Resolves technical and scope conflicts by gathering the facts first, logging open decisions, and getting sign-off.

Work Experience

Software Engineer
LvtGames
Aug 2025–Mar 2026 · Istanbul, Türkiye · Remote · Full-time
- Built high-throughput transaction processing and multi-balance wallet services in Go 1.24 and Spring Boot 3 (Java 21) for a multi-tenant B2B2C entertainment platform.
- Provisioned GCP infrastructure with Terraform, implemented Bitbucket layer-cached CI/CD pipelines that cut deployment times by ~30%, and led cross-functional feature launches.

Tools: Go 1.24, Java 21, Spring Boot 3, PostgreSQL, Redis, GCP, Terraform, Liquibase

Software Engineer
SwapWallet
May 2025–Jul 2025 · Tehran · Full-time
- Resolved critical ledger concurrency race conditions and optimized the crypto exchange matching engine under peak load, reducing p99 execution latency.
- Automated administrative and member operations by designing an event-driven Telegram workflow bot integrated with core exchange services.

Tools: Kotlin, Spring Framework, Apache Kafka, PostgreSQL, Kubernetes

Software Engineer
Ultratrader
Feb 2024–Oct 2024 · Toronto, Canada · Remote · Full-time
- Designed a real-time ingestion pipeline that streams and normalizes financial market data across 30+ external broker APIs at ~1,000 events/sec sustained, with bursts to 300k events/sec.
- Built REST and GraphQL APIs alongside low-latency broker proxy gateways, and optimized analytical PostgreSQL queries to cut reporting latency by more than 90%.

Tools: NestJS, TypeScript, PostgreSQL, Redis, GraphQL, Docker

Software Engineer
Ravindex
May 2022–Dec 2023 · Tehran
- Ingested and processed stock-market tick data through high-throughput analytical pipelines backed by replicated PostgreSQL databases.
- Developed event-driven microservices using Django and RabbitMQ across 20+ automated data-fetching workers, instrumented with Prometheus and Grafana alerting.

Tools: Python, Django, PostgreSQL, RabbitMQ, Docker, Prometheus, Grafana

Research and Development Intern
Plotset
Aug 2020–Jul 2021 · California, US · Remote
- Researched AI-driven data visualization and architected a modular rendering framework integrating multiple charting engines in Node.js and D3.js.

Game Developer
CubeCraft
Sep 2016–Feb 2017 · London, UK · Remote
- Built customizable and maintainable Spigot game mode plugins handling high concurrency.
- Designed state machines with 50+ states and 100+ transitions to manage complex gameplay loops.
- Implemented over 300 event listeners and schedulers to optimize real-time tick performance.

- Debugged and enhanced low-level libraries in the core Java codebase.

Tools: Java, Spigot API, PaperMC, MongoDB

Other Experience

Developer & Technical Product Manager
Manova: Falak
Jul 2026–Present · Sari, Mazandaran, Iran

- Designed and built Falak, a multi-tenant SaaS for restaurant operations covering stock across locations, receipts with partial payments and overdue tracking, and role-based access with location-scoped roles.
- Ran requirement sessions with restaurant owners who were not technical, turning their daily routines into module specifications, acceptance flows, and architecture decisions that set each release's scope.
- Defined and verified the UX rules for staff working on phones and tablets: 390 px and 768 px checked before desktop, 44 px touch targets, full-screen flows on small screens, and Persian and English in RTL at every width.
- Modeled stock as an append-only movement ledger with row-locked atomic batches and negative-balance guards, and enforced tenant isolation with PostgreSQL row-level security (FORCE RLS) on the inventory, receipts, and identity tables.

Tools: Go, PostgreSQL, Next.js, Orbit Auth

Developer & Technical Product Manager
Goldstein
Jul 2026–Present · Mashhad, Khorasan, Iran · Contract

- Co-built a multi-tenant SaaS platform for jewelry retailers to trade physical assets in real time, covering customer verification, a digital wallet ledger, live buy and sell pricing, and physical delivery.
- Ran requirement sessions with store owners who were not technical, converting them into a feature catalog, state machines, and a logged list of open decisions, with a written change process so scope moved only with sign-off.
- Built a concurrency-safe trading engine with pessimistic row locks on inventory and wallet balances, no short selling, and an append-only audit ledger for every balance change.
- Load-tested the engine and traced a cross-store slowdown (one busy store slowed others about 33× at p50) to a shared database connection pool. Sizing the pool and adding PgBouncer brought the slowdown to 1.35× in tests.

Tools: NestJS, Next.js, PostgreSQL, Prisma, Redis, PgBouncer

Head Teaching Assistant — Advanced Programming
University of Tehran
2026–Present · Tehran

- Lead a team of 15+ TAs across problem design, code reviews, grading, and curriculum delivery for 50+ undergraduate students per semester.
- Designed and maintain the containerized Docker CI judge and automated assignment evaluation toolchain used across all terms for Java OOP, SOLID, and concurrency coursework.

Co-Founder & Lead Architect
Hambazi — Boardgame & Startup Community
Aug 2025–Present · Tehran

- Built a Telegram bot for user, event, and game management, backed by the community platform backend (NestJS and Next.js) that supports event scheduling, matchmaking, and member workflows for 1,000+ active members.

Education

M.Sc. in Computer Engineering — Algorithms and Computation
University of Tehran · Sep 2024–Present

Research focus: Large language models, hallucination, and mechanistic interpretability. Thesis: Interpreting Neural Network Thoughts.

B.Sc. in Computer Engineering
Babol Noshirvani University of Technology · Sep 2018–Aug 2024

Thesis: High-Throughput Distributed UUID Generation System — Engineered a collision-free 128-bit (16-byte) distributed ID generator in Go and RabbitMQ sustaining 10,000,000 IDs/sec per node with automated node coordination.

Projects

Promux
Autonomous multiplexer for agentic AI CLIs with zero-downtime account rotation.

Architected a proxy multiplexer that intercepts agentic CLI calls. Automatically detects HTTP 429/500 quota exhaustion, triggers account rotation with TTL cooling, and preserves running agent session context without interrupting long code generation loops.

Go, CLI, AI Systems

Manova / Orbit Platform
Shared foundation for client products: auth, limits, billing, and tenant isolation.

Shared micro-platform for client products, with RBAC, token-bucket rate limits, billing journals, and partitioned PostgreSQL for multi-tenant isolation across 5+ applications.

TypeScript, Go, PostgreSQL, Edge

Dorost

Open-source bilingual document composition package on official Typst registry.

Authored and published an open-source Typst package providing zero-dependency RTL/LTR mixed-direction layout, numeral conversion, and design system components with automated CI test suites.

Typst, Open Source, Design System